rsera | WHIZLABS

0/3 learning items

Fundamentals of Deep
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Data is processed in a Neuron ?
• 5 min

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• 8 min

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• 8 min

Learning Neural Network - Forward
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• 4 min

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• 4 min

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• 6 min

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• 8 min

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dations of Deep Learning - Assessment
ed Assignment • 20 min

l Deep Learning Techniques

Question

Which activation function is similar to sigmoid but maps the input to a range between -1 and 1?

- Sigmoid
- Hyperbolic Tangent (tanh)
- Rectified Linear Unit (ReLU)

Correct
Feedback: This is Correct. The tanh function is similar in shape to the sigmoid but its output range is between -1 and 1. This centering often leads to faster convergence during training compared to sigmoid For more information, refer to the video "Activation Functions - Demo".

Continue | Skip

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But in the case of tanh, the new range is minus 1 to plus 1

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Activation Functions - Demo

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ndamentals of Deep

of Advanced Deep Learning Techniques
10 min

s Classification with MNIST Dataset -
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3 min

Multiclass Classifier - Fit and Evaluate
min

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min

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Demo
min

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l Deep Learning Techniques -
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ssignment • 15 min

ways of the course
10 min

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10 min

Question

What is the primary purpose of using softmax activation in the final layer?

- To produce output values that sum to 1, representing probabilities for each class
- To introduce non-linearity
- To scale the output values between 0 and 1
- To ensure all output values are positive

Correct
Feedback: This is Correct. Softmax converts the raw output scores into probabilities, ensuring they are non-negative and sum to 1, which is ideal for multi-class classification. For more information, refer to the video "Training Multiclass Classifier - Fit and Evaluate".

Continue | Skip

that's great. Now the best part is whenever we create any model all these

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Training Multiclass Classifier - Fit and Evaluate

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NVIDIA: Fundamentals of Deep Learning

- How Data is processed in a Neuron ?
  Video • 5 min
- **Gradient Descent**
  Video • 8 min
- Training a Perceptron - Demo
  Video • 8 min
- Deep Learning Neural Network - Forward Propagation
  Video • 4 min
- Backward Propagation - Deep Learning Neural Network
  Video • 4 min
- Activation Functions
  Video • 6 min
- Activation Functions - Demo
  Video • 8 min
- Introduction to Deep Learning & Neural Networks - Knowledge check
  Practice Assignment • 15 min
- Foundations of Deep Learning - Assessment
  Graded Assignment • 20 min

Module 2
Advanced Deep Learning Techniques

Question

Which gradient descent variant calculates the gradient using the entire training dataset in each iteration?

- Batch Gradient Descent (BGD)
- Stochastic Gradient Descent (SGD)
- Mini-Batch Gradient Descent (MBGD)

Correct
Feedback: This is Correct. BGD computes the gradient using all training examples, leading to accurate updates but high computational cost for large datasets. For more information, refer to the video “Gradient Descent”.

Continue Skip

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Now while using this gradient descent technique on a given dataset we are going to face some

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Gradient Descent

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